GTC4Lusso USA - Technical specifications - Latest Update: 2021-01-26

C1.01 Technical specifications

DCT gearbox

Dual clutch transmission, 7 forward gears + reverse, electrohydraulic system controlled automatically or in semi-automatic mode from the steering wheel shift paddles.

Each of the two clutches is associated to a different part of the gearbox, with one part engaging even gears and the other engaging odd gears.

Each time the driver selects a gear, the system already preselects the next gear.

As the correct engine speed is reached, one clutch opens while the other closes simultaneously, so that the shift is performed with no interruption in power delivery.

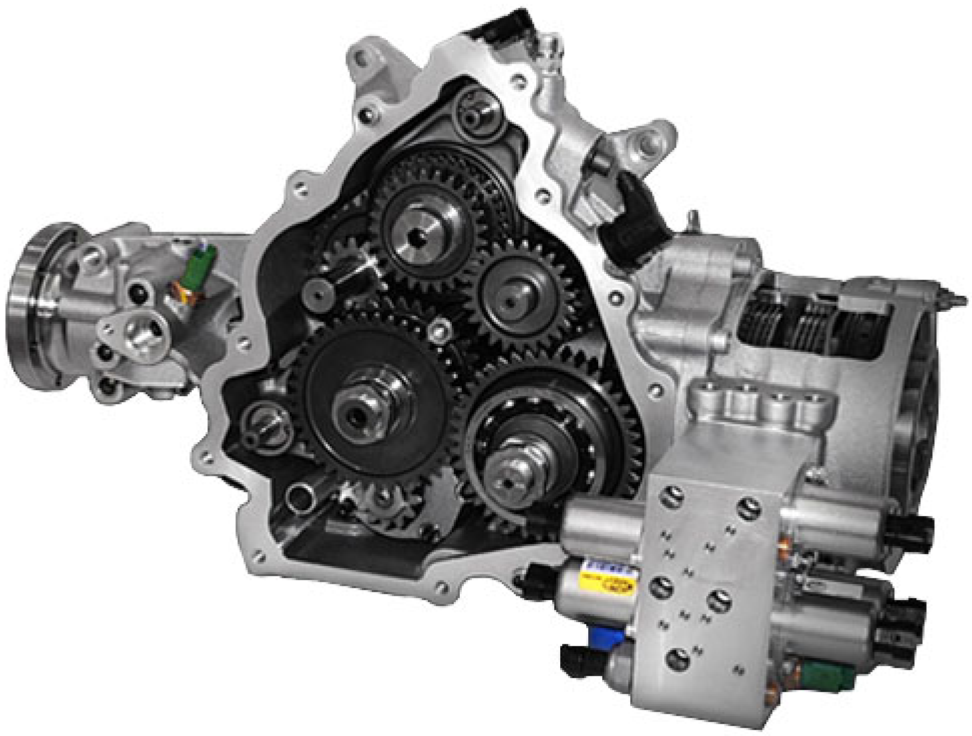
Electronic control systems

E-Diff: active electronic differential on rear axle, which continuously varies the degree of differential lock-up between the two axle shafts.

Gear ratios

	1st	2nd	3rd	4th	5th	6th	7th	R
Transmission gear ratios	3.397	2.185	1.626	1.286	1.028	0.839	0.634	2.791
Bevel gear pair/differential ratio			3.875					

4 wheel drive system



This vehicle is equipped with an innovative four wheel drive system developed by Ferrari, which transfers torque to the front wheels only in specific low grip conditions.

Unlike a conventional all-wheel drive system applied to a front engined car, this solution maintains the traditional Ferrari powertrain architecture with a front-mid engine and a rear transaxle connected to the engine by a single drive shaft.

Torque, modulated by the new Ferrari Dual Drive electronic control management system, is transferred to the front wheels via the Power Transfer Unit (PTU), a compact transmission system located on the front axle and connected directly to the crankshaft.

This mechanical configuration:

- weighs 50% less than a conventional all-wheel drive system, with evident advantages in terms of power/weight ratio and, as a result, performance.
- keeps the centre of gravity of the vehicle low and does not alter the typical rear-biased weight distribution of a Ferrari, with over 50% (53%) of the total weight of the vehicle over the rear axle even in a front-engined car.

As well as ensuring the correct drive ratio to match engine speed and wheel speed, the PTU also manages the torque transmitted to the front wheels, distributing the torque as necessary between the right and left wheels.

The PTU receives drive and torque directly from the crankshaft, and transmits this drive and torque to the front wheels via a system of gears and a pair of continuously slipping multiplate wet clutches with carbon discs.

As a result, the rear and front axles are not mechanically constrained to each other, and are connected instead to two entirely independent transmissions, allowing the vehicle to function in pure rear-wheel drive mode without sacrificing any of the typical driving pleasure of cars with a rear-drive architecture.

The multiplate discs compensate for differences between the front and rear wheels but also for differences in wheel speed across the front axle itself, fulfilling the same purpose as both the central and front differentials in a conventional 4WD system.

The front clutches are also completely independent, and are capable of transmitting different quantities of torque to the right and left wheels for implementing torque vectoring strategies.

The torque transmitted to the wheels is controlled by closing and opening the disc packs of each clutch independently.

The PTU is therefore extremely compact (measuring just 170 mm in length) and light, resulting in a 4WD system 50% lighter than a conventional system.

Like the F1 Dual Clutch gearbox, the hydraulic actuator system for the gears and clutches is integrated directly in the PTU, ensuring extremely rapid actuation times and faster system responsiveness.